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Note:

Component parameter description

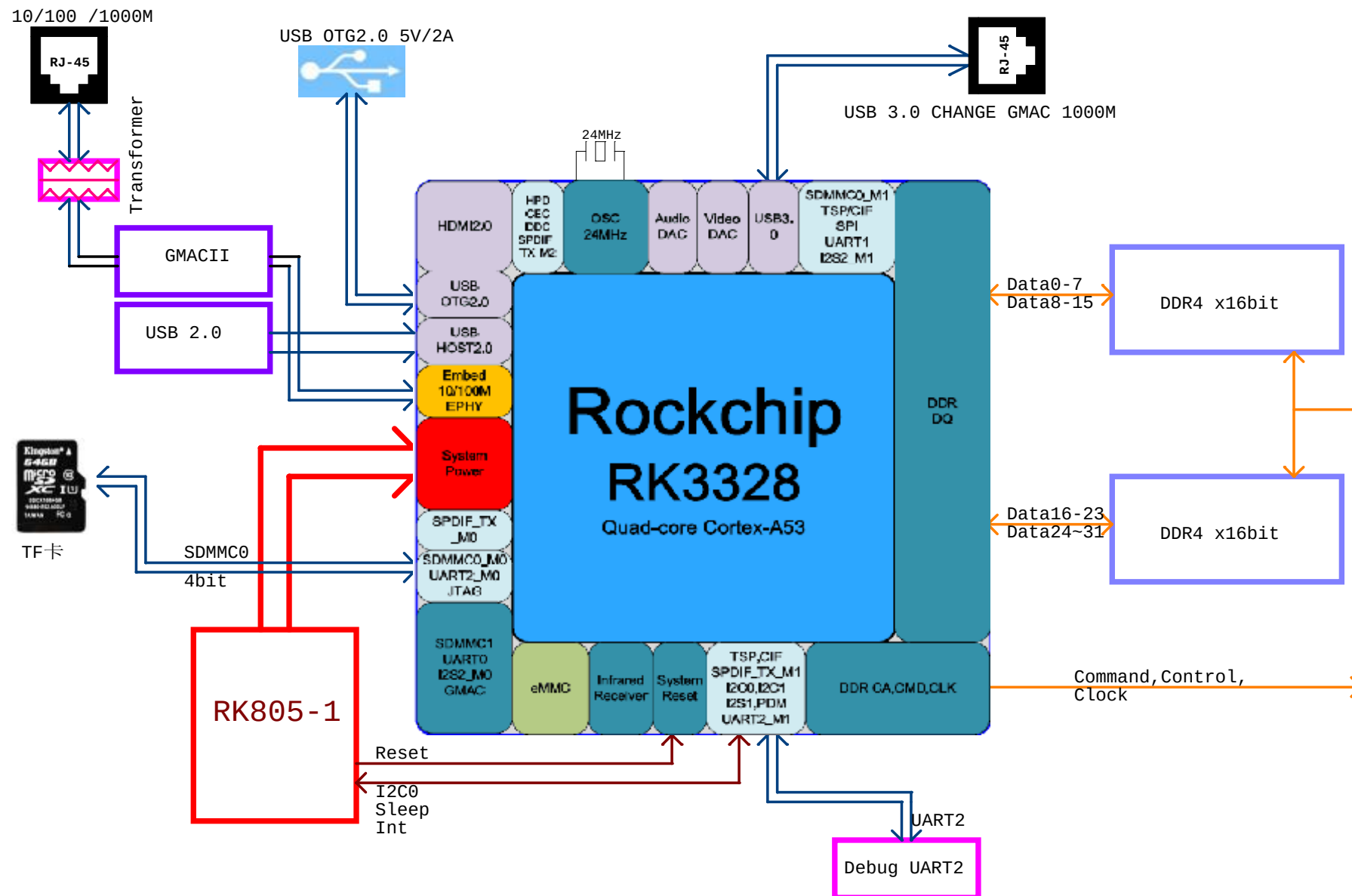
1. DNP stands for component not mounted temporarily

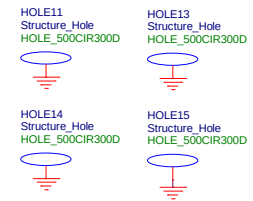
2. If Value or option is DNP, which means the area is reserved without being mounted

			
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Version	Date	Author	Change Note	Approved
V1.0	20170718	ZDZ	First edictor	
V1.0	20200509	DCL	1. Page5, 增加led指示模块, KEY-RESET, FAN模块; 2. Page9, 增加EEPROM WITH MAC ADDRESS模块, SPI-FLASH模块; 3. Page13, 14, 网口座子增加ESD器件, 网口两指示灯增加上拉电阻; USB3.0的RX, TX与CPU管脚顺序对调; 4. Page12, CON12座子网络定义同ZERO (1组I2C, 1组UART, LINEOUTR-L, TV-OUT), MIC没有导出来; PAGE10 GMAC_TXCLK修改, HOLE15接PGND, OTG接口增加ESD器件, 去掉HDMI1V8电源。 5. 增加HDMI1V8电源, PMU处晶振封装修改, RTL8153B 的RX, TX顺序交换; DC5V供电座子修改成TYPE-C接口。	
V1.1	20200811			
V1.2	20200929		6. USB2.0座子封装修改, 工厂容易贴片; R906 NC, R908贴片;	
V1.3	20210428		7. 更改RTL8211E为RTL8211F 8. RK3328 SD卡总线部分功能模块增加1.8/3.3V电源切换电路 9. 删除RK805-1预留的32.768K晶振部分电路, 网口指示灯处增加两个4.7K下拉电阻;	



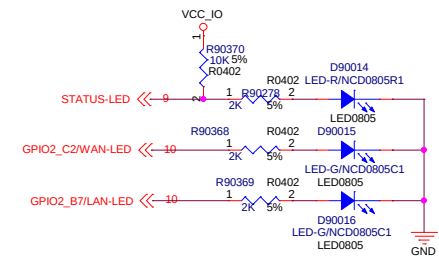




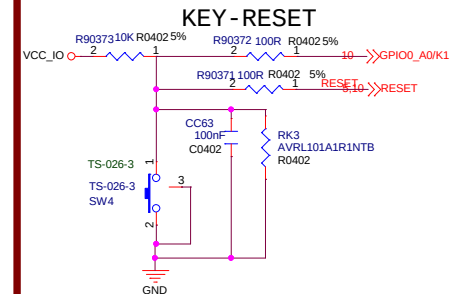
IF TVS UNMOUNTED,
ESD OR SURGE SHOULD BE
DAMAGE THE PMIC!!!

- 1: 如WIFI/BT需要32.768KHZ时钟，默认由CPU提供32.769KHZ时钟，默认不贴。
- 2: 如果修改了PLL分频，造成无法分出32.769KHZ时钟，然后WIFI/BT又需要32.768KHZ时钟，那么这部分电路需要贴上。
- 3: WIFI/BT不用32.768KHZ时钟，这部分可以不贴。

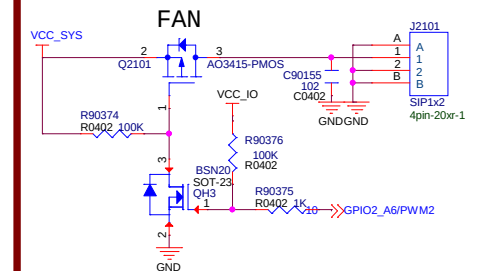
LED DISPLAY



KEY-RESET

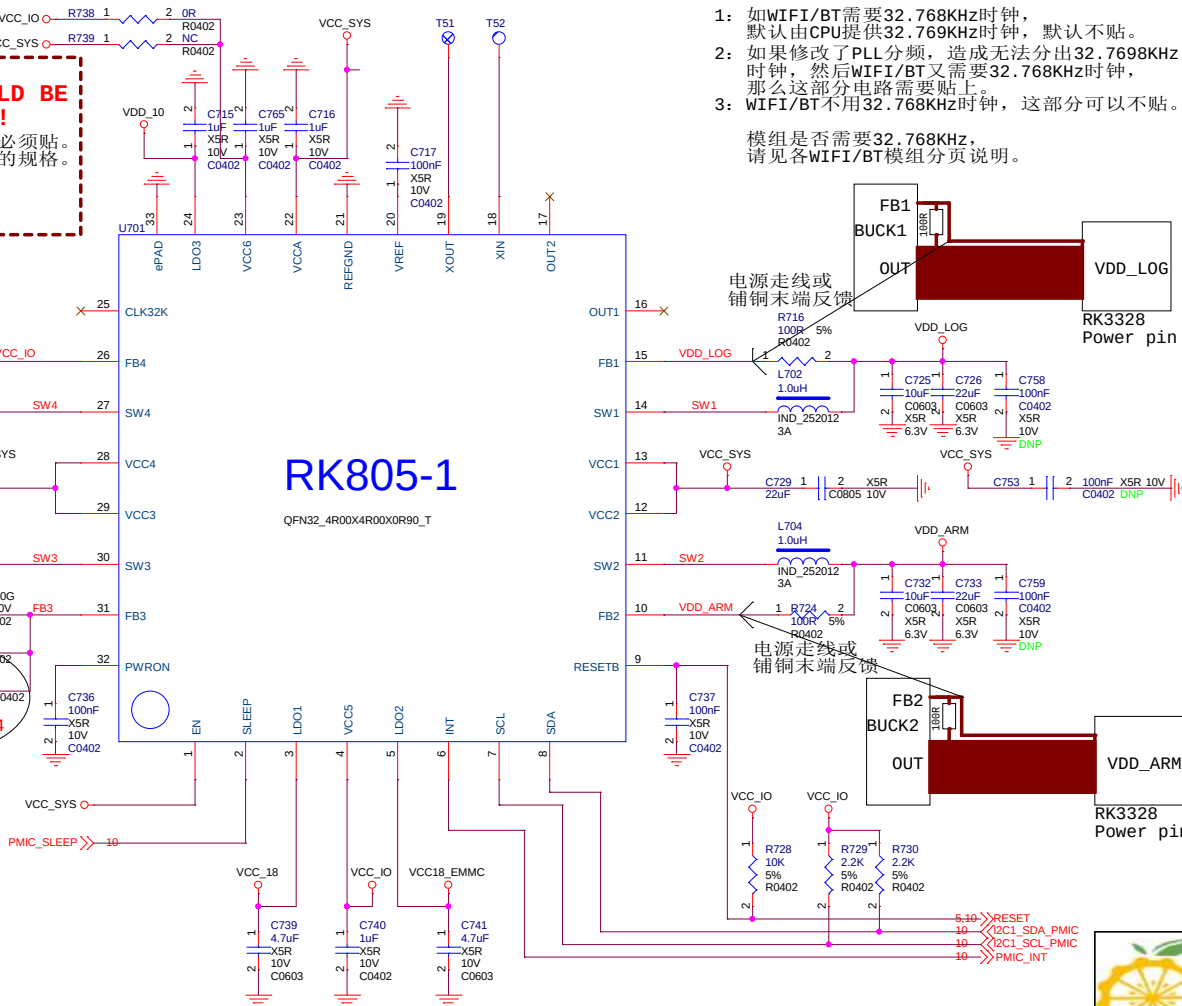


FAN



VCC DDR

	DDR3	DDR3L	DDR4/LPDDR3
VCC_DDR	1.527V	1.353V	1.2V
R725	100K 1%	47K 1%	51K 1%
R726	110K 1%	68K 1%	100K 1%



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Name
System Power-PMIC RK805

C

WIC

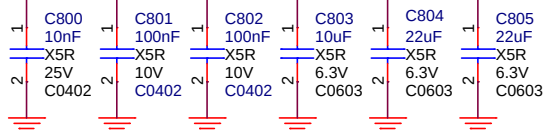
1

U800N

ARM

VDD_CORE
VDD_CORE
VDD_CORE
VDD_CORE
VDD_CORE
VDD_CORE
VDD_CORE

P13
P14
R13
R14
T13
T14
T15

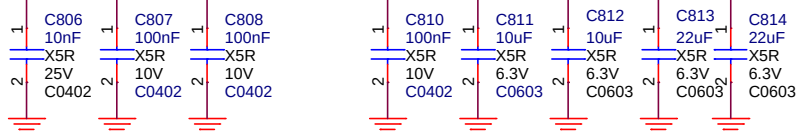


PCB LAYOUT注意: 电容都要靠近主控, 其中一个22uF电容需要放在管脚正下方。

GPU/Logic

VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC
VDD_LOGIC

N9
N10
N11
P9
P10
P11
R9
R10
R11

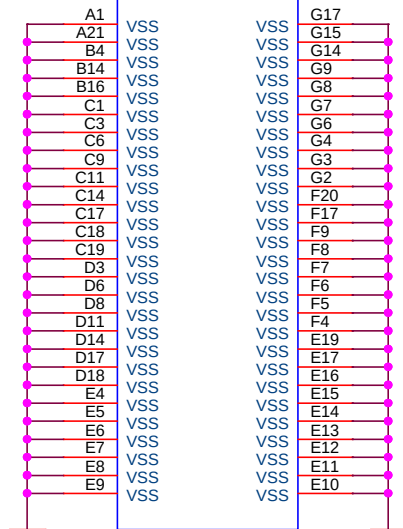


PCB LAYOUT注意: 电容都要靠近主控, 其中一个22uF电容需要放在管脚正下方。

RK3328

BGA395_14R00X14R00X1R24

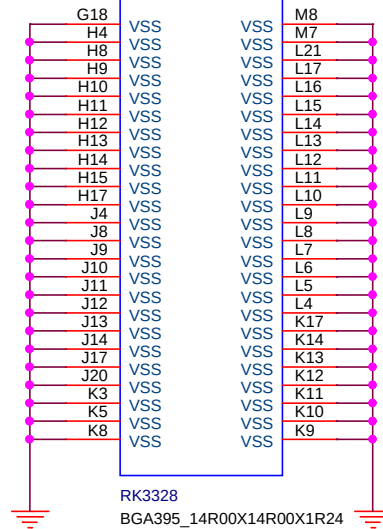
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RK3328

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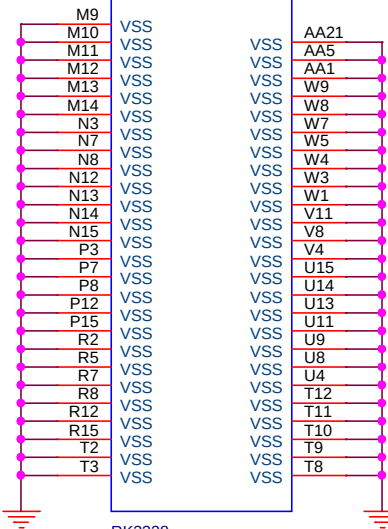
U800P



RK3328

BGA395_14R00X14R00X1R24

U800Q



RK3328

BGA395_14R00X14R00X1R24

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U800A

OSC

XOUT24M

XIN24M

PLL Power

PLL_DVDD_1V0

PLL_AVDD_1V8

SARADC

SARADC_IN0

SARADC_IN1

SARADC_AVDD_1V8

OTP/eFUSE

OTP_VCC18

EFUSE_VP

RK3328

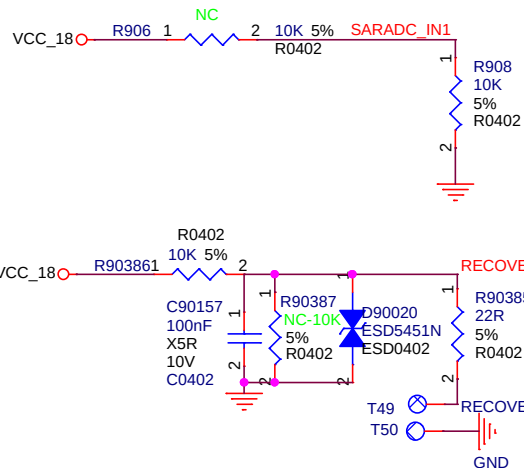
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VDDPLL/USB30_1V0

VDD_10

VDDPLL/USB30_1V8

VCC_18



SARADC_IN0 BOM版本	SARADC_IN1 PCB版本	Up阻值	Down阻值	
Recover		DNP	10K	
BOM1		110K	10K	
BOM2		100K	20K	
BOM3		100K	33K	
BOM4		36K	18K	
BOM5		51K	36K	
BOM6		51K	51K	
BOM7		36K	51K	
BOM8		18K	36K	
BOM9		33K	100K	
BOM10		20K	100K	
BOM11		10K	110K	
BOM12		10K	DNP	

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Design Name

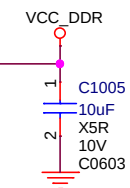
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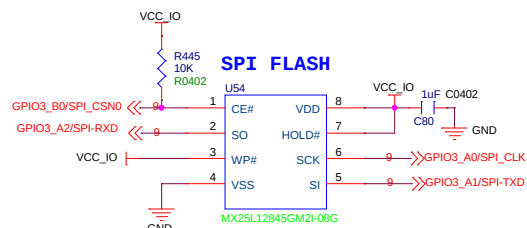
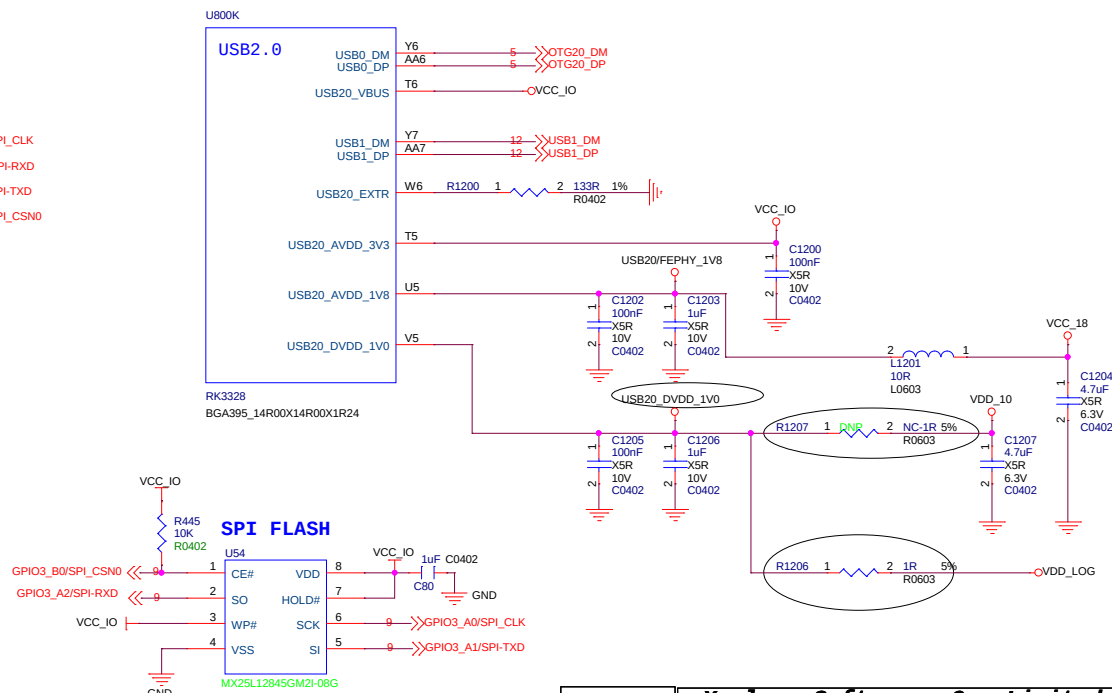
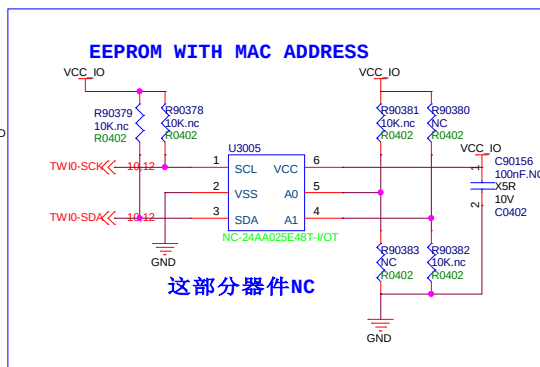
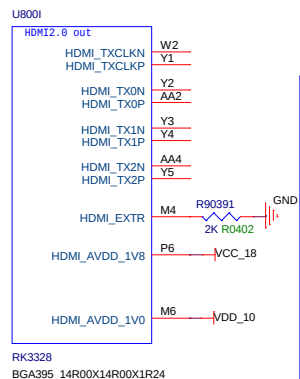
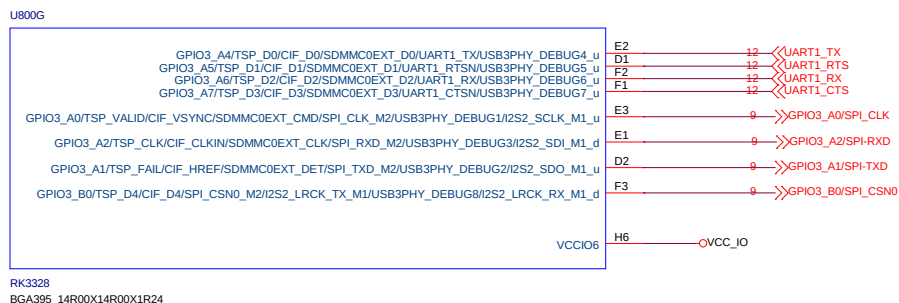
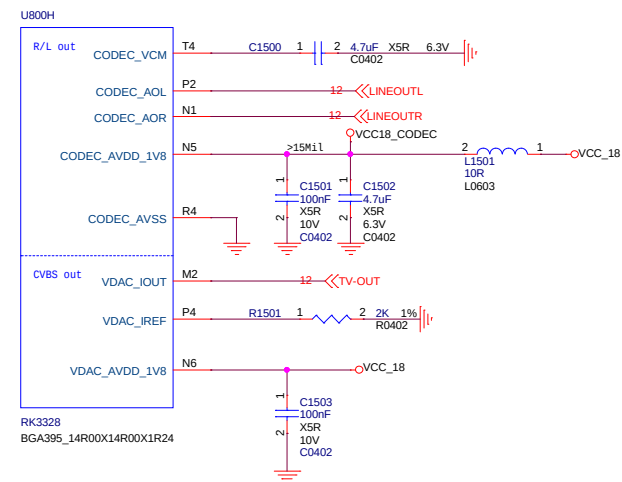
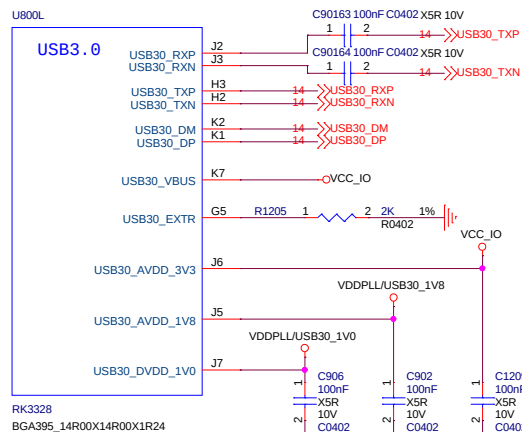
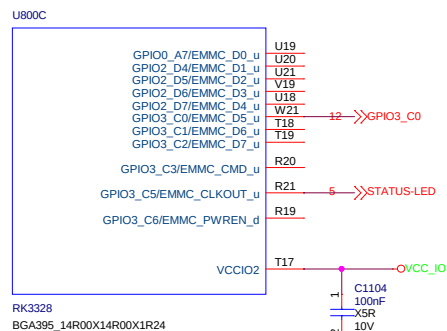
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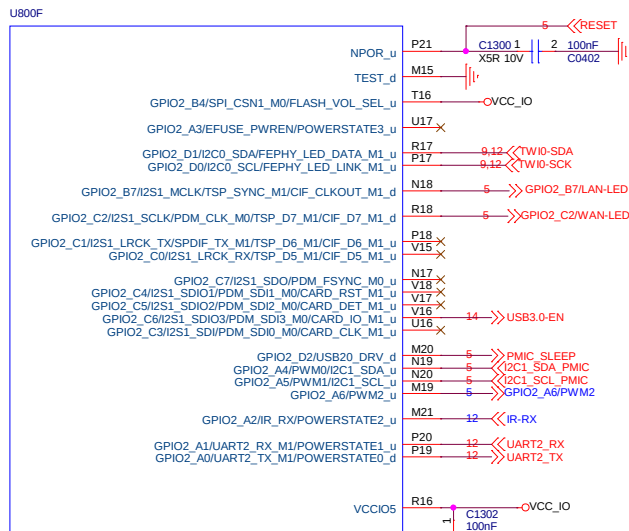
OSC/PLL/OTP/SARADCRev
1.2

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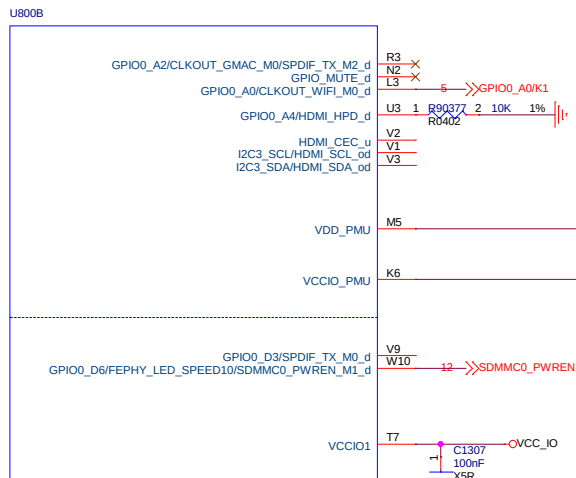
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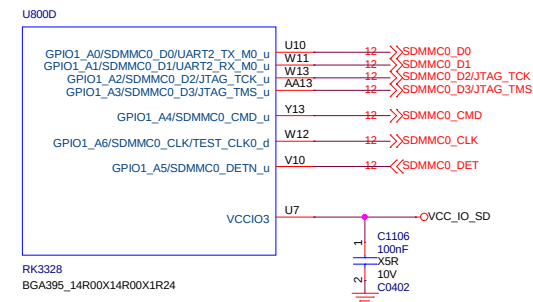




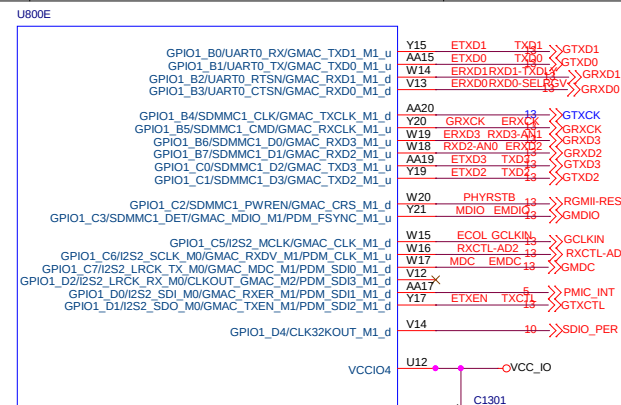
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BGA395_14R00X14R00X1R24



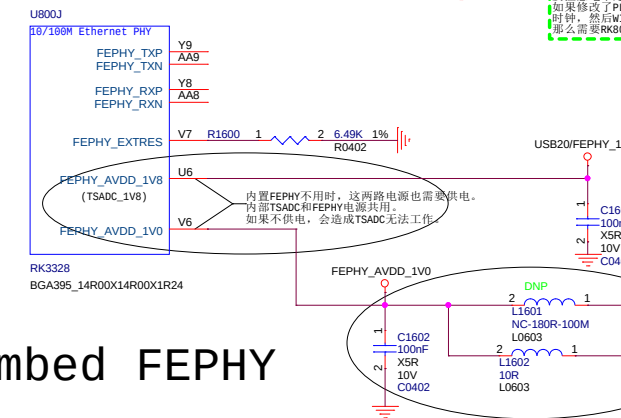
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RK3328
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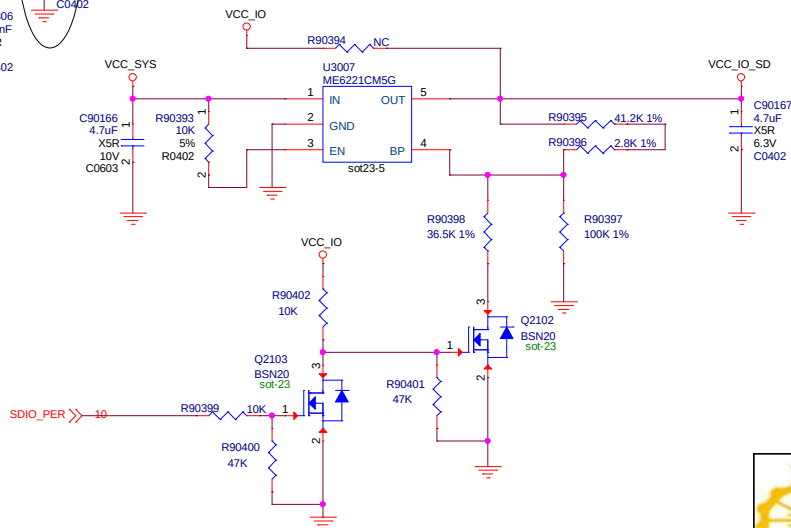


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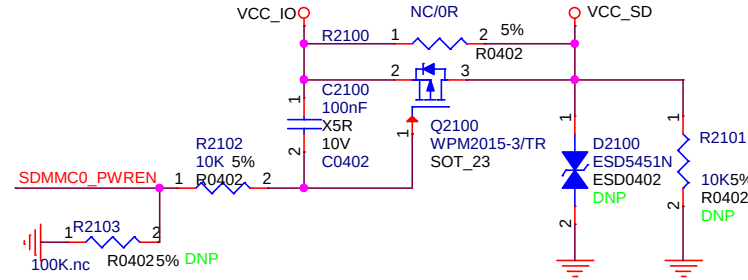
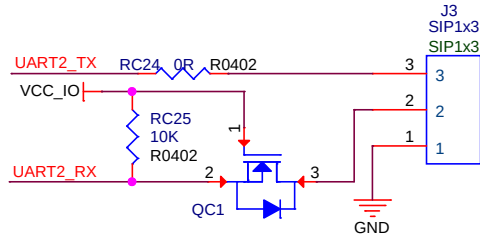
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Embed FEPHY

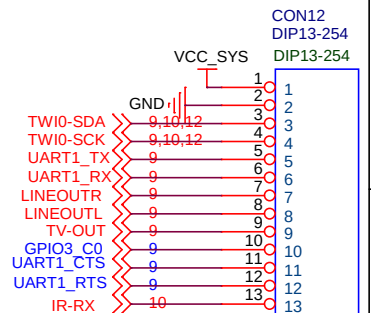
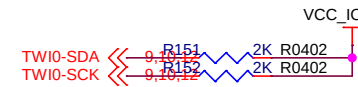
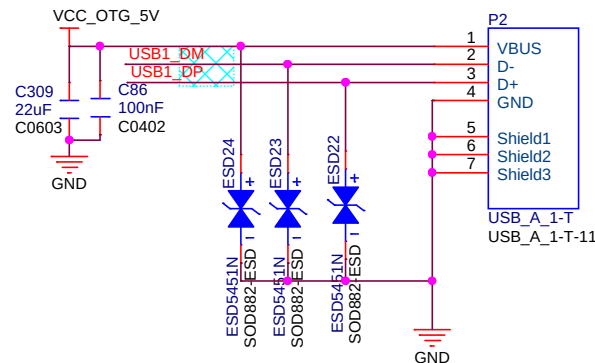
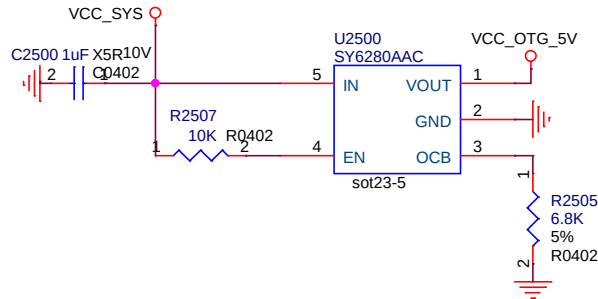
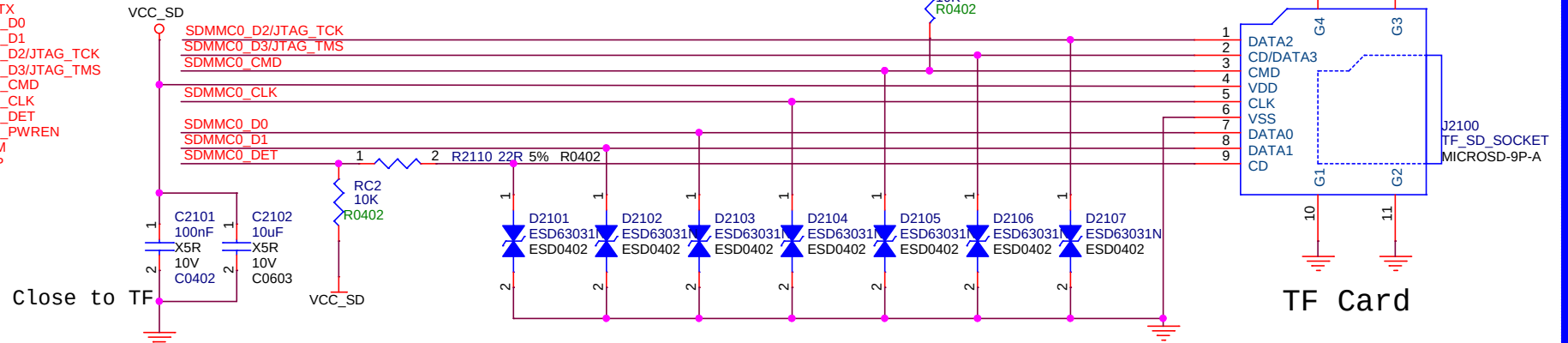


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Debug UART2



10 >> UART2_RX
 10 >> UART2_TX
 10 >> SDMMC0_D0
 10 >> SDMMC0_D1
 10 >> SDMMC0_D2/JTAG_TCK
 10 >> SDMMC0_D3/JTAG_TMS
 10 >> SDMMC0_CMD
 10 >> SDMMC0_CLK
 10 >> SDMMC0_DET
 10 >> SDMMC0_PWREN
 9 >> USB1_DM
 9 >> USB1_DP



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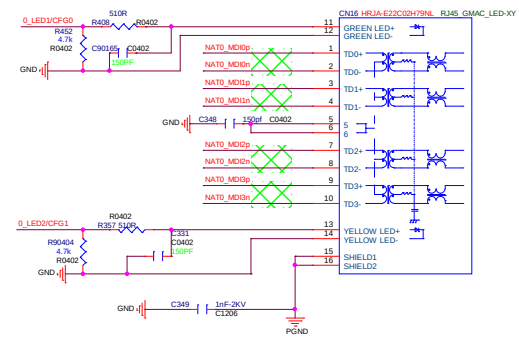
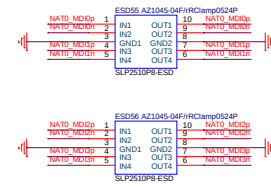
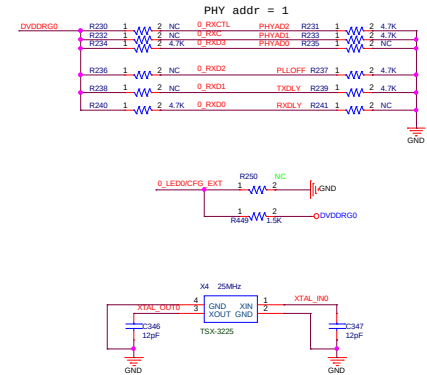
TF/SD Card/debug/usb2.0

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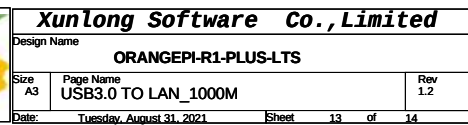
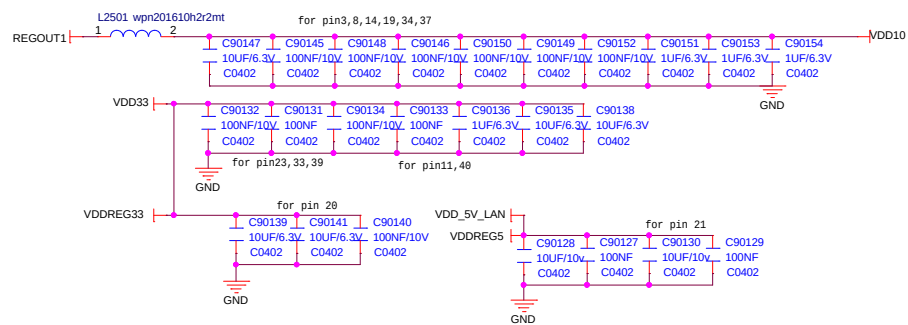
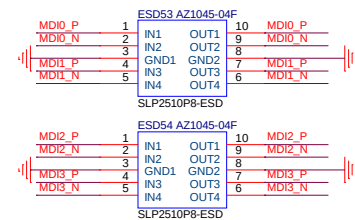
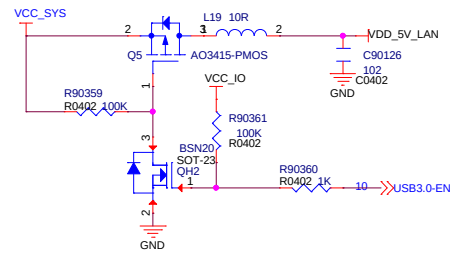
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10/100/1000 MII Ethernet PHY



LED2:Blinking=Transmitting or Receiving.
LED1:Link Up/Down



LPDDR3 >=1866Mbps

